

PORTFOLIO HEALTH & EXECUTIVE OVERVIEW

Macro origination trends, capital allocation, and top-level portfolio solvency metrics.

Academic Case Study: Built with anonymized secondary sample data for educational demonstration
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Region

All

Total Applications

149K

Total Disbursed Volume

49bn

Default Rate %

24.64%

Default Exposure \$

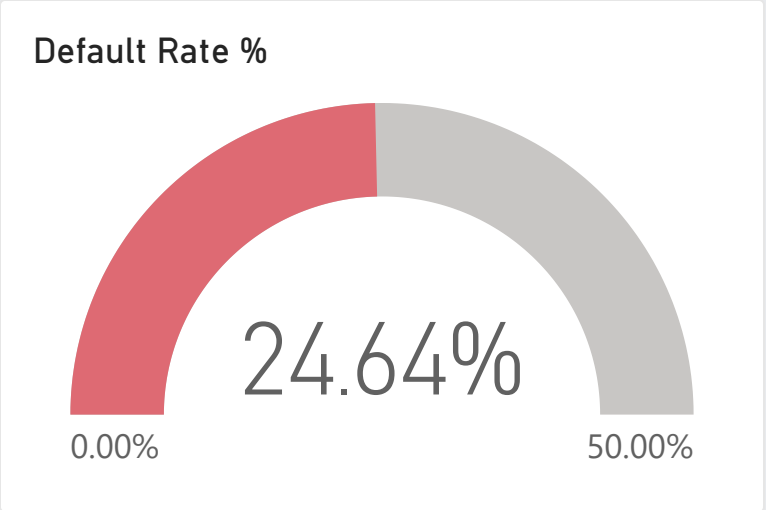
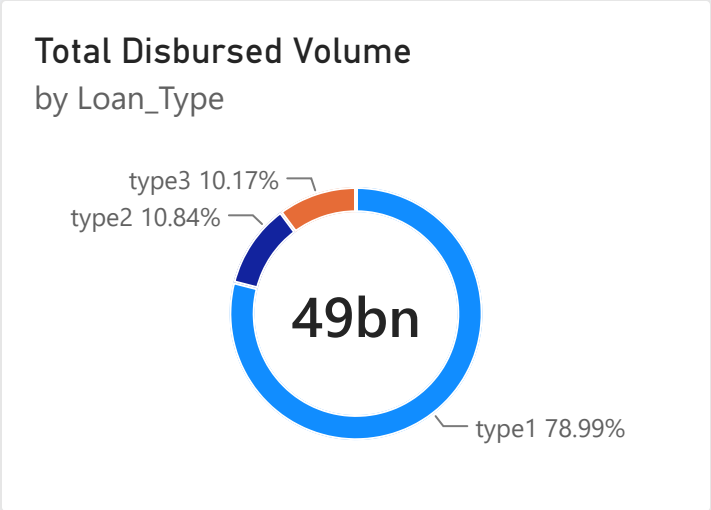
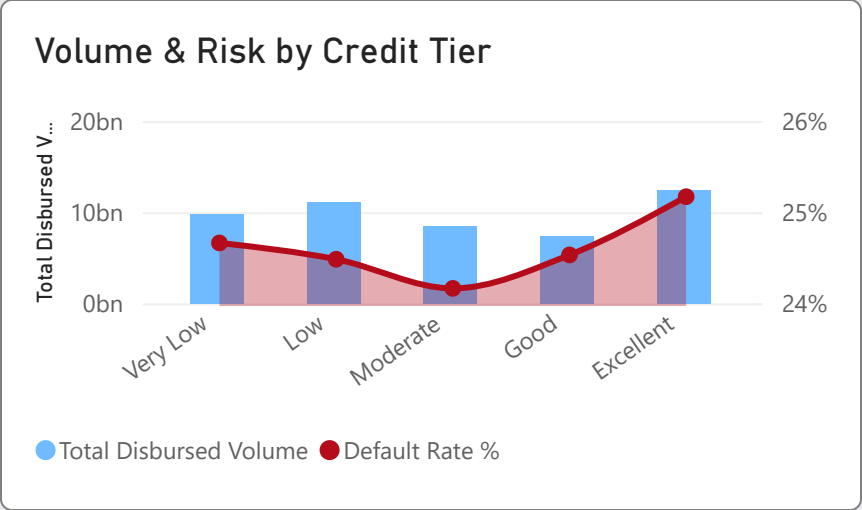
\$11.70bn

Avg Interest Rate

4.03%

High Risk Exposure \$

2bn



Portfolio Risk Summary: Total defaulted exposure sits at \$11.70bn (24.64% rate), heavily driven by 'type 1' loans. Regional hot-spotting indicates significant capital at risk. Immediate recalibration of underwriting thresholds is recommended to preserve capital. Data reflects a static 2019 sample snapshot | Currency: USD | N = 149K applications.

ORIGINATION & UNDERWRITING ANALYSIS

Examine portfolio origination quality, regional leverage DTI and disbursement volume.

Region

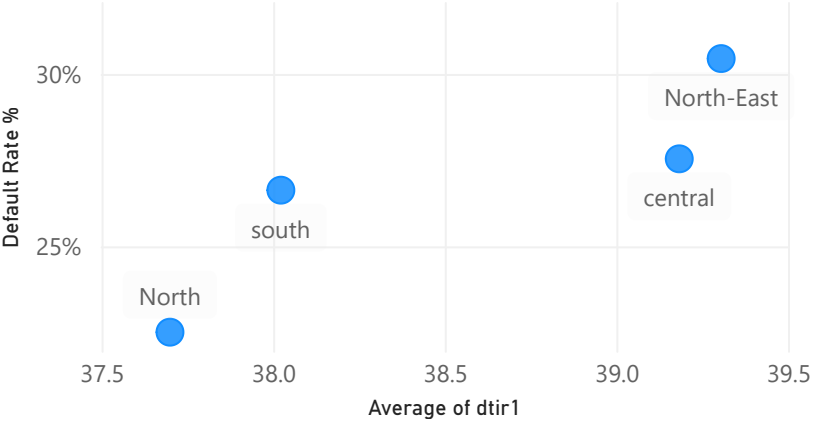
All

Loan_Type	Total Applications	Avg Loan Amount	Avg Interest Rate	Default Rate %
⊕ type1	113173	\$3,43,587	4.09%	22.77%
⊕ type2	20762	\$2,56,942	3.98%	34.54%
⊕ type3	14735	\$3,39,863	3.70%	25.06%
Total	148670	\$3,31,118	4.03%	24.64%

LOAN TERMINOLOGY

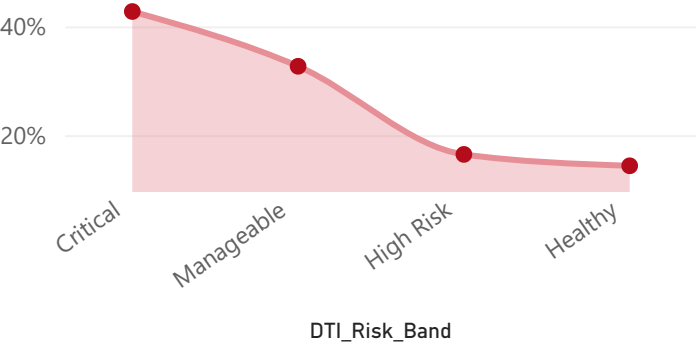
- Type 1 (Unsecured): Standard personal credit. Primary driver of total origination volume.
- Type 2 (Auto/Asset): Short-term secured lending. Currently peaking in default frequency.
- Type 3 (Prime/Real Estate): Long-term secured debt. Most stable baseline performance.

Default Rate % vs Average DTI by Region



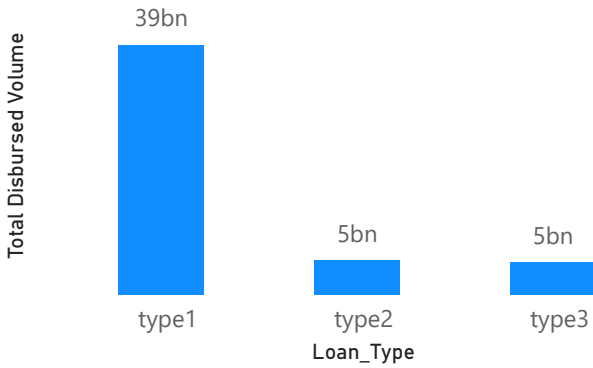
Delinquency by DTI Band

Default Rate % by DTI_Risk_Band



Disbursement Bridge by Loan Type

Total Disbursed Volume by Loan_Type



Executive Takeaway: Restrict Type 1 and Type 2 originations within 'Critical' DTI bands; immediately reallocate funding toward high-yield, stable Type 3 secured instruments.

RISK & DEFAULT DRIVER ANALYSIS

Region

All

Identify and concentrate attention on the highest-exposure segments

Key influencers Top segments

What influences Status to ?

When...
Note: 'Status' indicates a defaulted loan.

...the average of Status increases by

LTV is 0 or less

0.84

LTV is more than 107.2033898

0.75

Income is 720 or less

0.75

dtir is more than 49

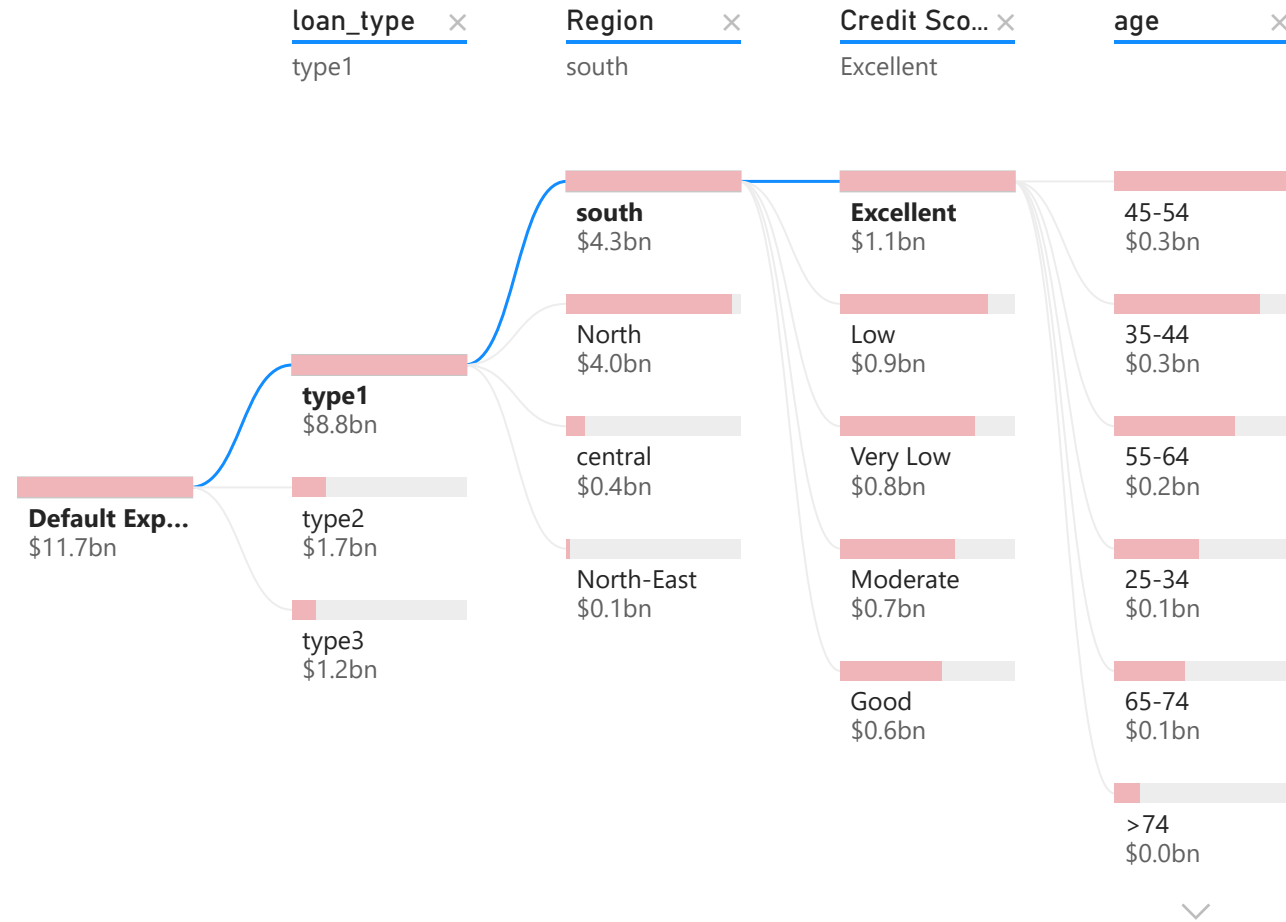
0.21

Income is 1020 - 2760

0.15

loan_type is type2

0.14



Why this matters: LTV above 107% is the single strongest predictor of default — this should be a hard underwriting cap, not just a risk flag.

PORTFOLIO EXPOSURE & WATCHLIST

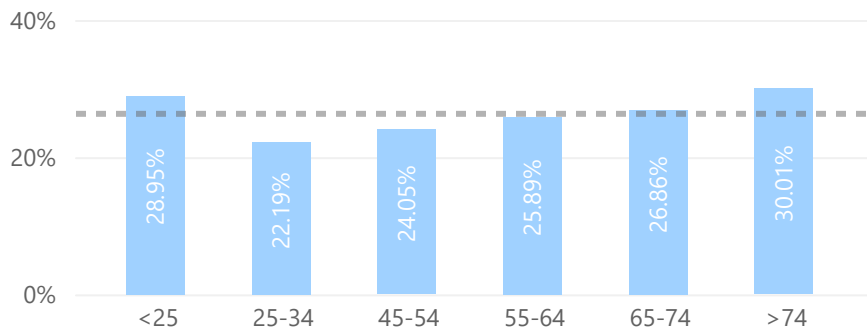
Region

All

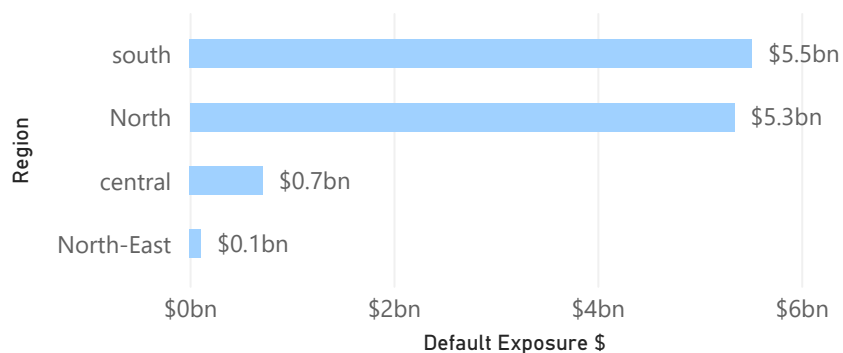


Track demographic risk thresholds, regional momentum, and top-exposure individual loans.

Default Rate % by Age Group

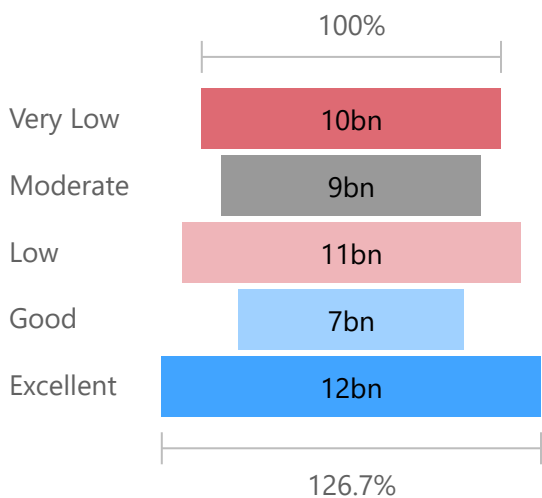


Default Exposure \$



Total Disbursed Volume

by Credit Score Band



High-Risk Watchlist

ID	Loan_Amount	Credit Score	LTV
29211	3576500	751	68.67
39495	2396500	869	104.29
42004	2476500	809	15.00
61414	2606500	526	56.56
67090	2926500	663	59.99
93311	2706500	767	60.04
109557	2406500	710	80.00
111298	3006500	511	85.70
160281	2276500	726	64.89
169975	2276500	885	64.89

Total

Action Plan: Data reflects the 2019 portfolio snapshot. Loan-to-Value (LTV) metrics exclude Type 1 (unsecured) credit lines.

Pricing, Solvency & Simulation

Region

All

Capital Adequacy Ratio

76.24%

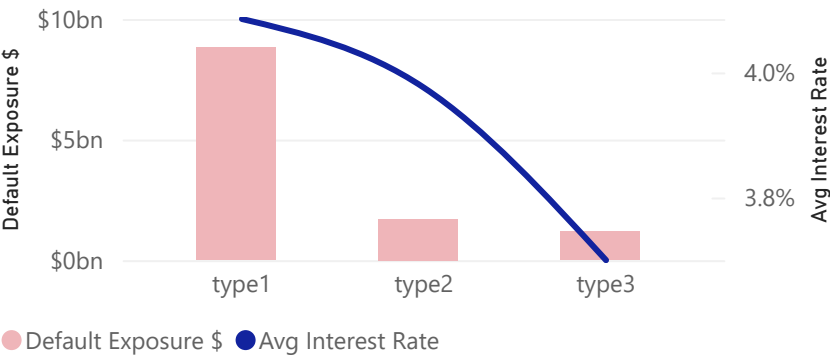
Capital Adequacy Ratio: Measures the proportion of disbursed capital remaining solvent after accounting for total defaults. The value of 76.24% that the remaining 23.76% of portfolio has experienced severe impairment.

Risk-Adjusted Yield

-9.71bn

Risk-Adjusted Yield: Compares gross interest income against total default exposure. The negative spread confirms that current pricing structures fail to price in default probability which results in net capital erosion.

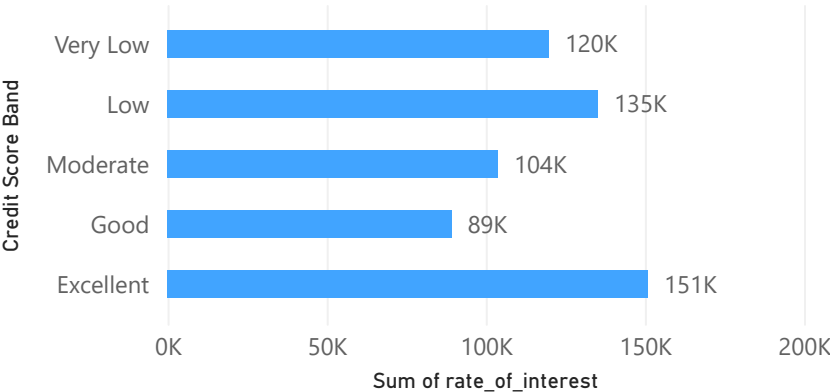
Default Exposure \$ and Avg Interest Rate by Loan_Type



Default Rate % by Region and Age Cohort

Region		<25	25-34	45-54	55-64	65-74	>74	Total
central		24.56%	33.33%	23.05%	29.03%	27.30%	31.14%	27.54%
North		19.88%	27.45%	20.21%	21.91%	24.06%	24.91%	22.51%
North-East		30.12%	46.67%	24.83%	27.10%	32.55%	33.12%	30.45%
south		25.52%	30.63%	24.57%	25.71%	27.65%	28.49%	26.63%
Total		22.74%	28.95%	22.19%	24.05%	25.89%	26.86%	24.64%

Sum of Rate Of Interest



Scenario Simulation & Stress Testing

Baseline (Current Portfolio)

Default Exposure \$
\$11.70bn

Risk-Adjusted Yield
-9.71bn

Capital Adequacy Ratio
76.24%

Simulated (Post-Adjustment)

Simulated Default Exposure
6.37bn

Simulated Risk-Adjusted Yield
-5.37bn

Simulated Solvency Rate
87.06%

LTV Cap Impact: Setting a strict LTV cap simulates an underwriting policy that eliminates the highest-risk borrowers from the portfolio, significantly reducing total default exposure at origination.

Yield Dynamics: Dropping interest rates decreases gross income, while stress scenarios increase baseline defaults. Observe how a tighter LTV cap can mathematically outpace those losses to protect net yield.

Capital Solvency: Enforcing a strict LTV cap directly raises the overall survival rate of the disbursed capital. This tool proves that tighter initial underwriting protects capital far better than adjusting interest rates after the fact.

Simulation Controls

LTV Cap Threshold

60



Interest Rate Adjustment

-0.02



Scenario

- ☐ Base
- ☐ Moderate
- ☐ Severe

Simulation Impact Breakdown

Region

All

Loan_Purpose

All

LTV Cap Threshold

60

Interest Rate Adjustment

-0.02

Scenario

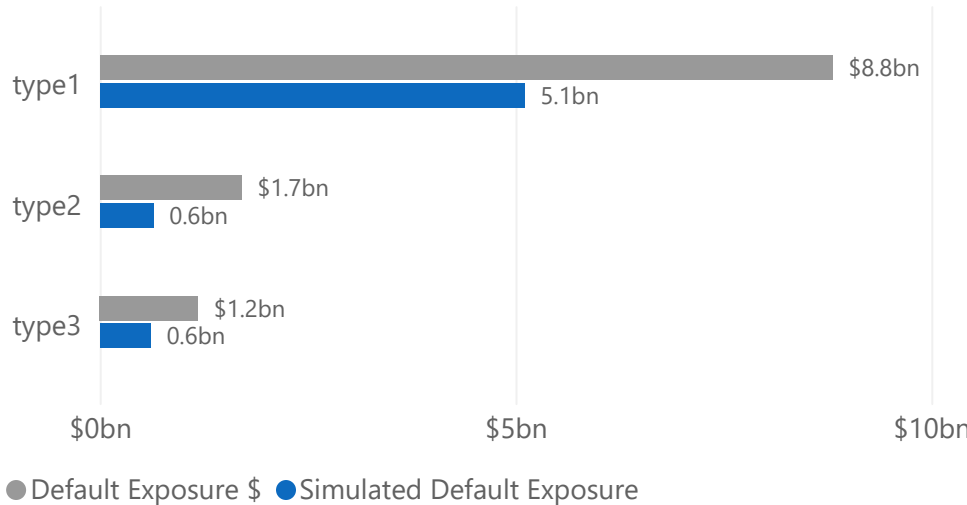
☐ Base

☐ Moderate

☐ Severe

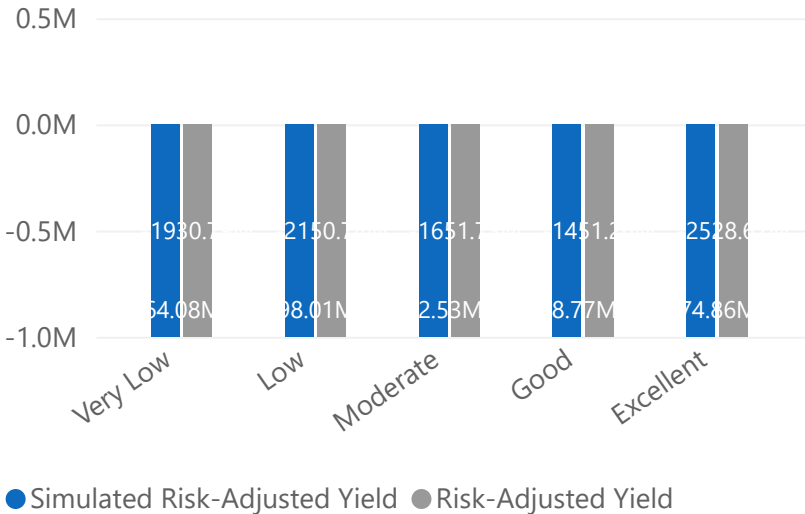
Exposure Impact

Default Exposure \$ and Simulated Default Exposure
by Loan_Type



Yield Impact

Simulated Risk-Adjusted Yield and Risk-Adjusted
Yield by Credit Score Band



Executive Note: Use the demographic dropdowns and simulation parameters on the left to dynamically identify which specific loan types and credit tiers carry the highest structural risk under economic stress.

Executive Portfolio Overview & Strategic Action Plan

- **Portfolio Health & Yield Deficit:** Current pricing structures fail to price in default probability, resulting in a 24.64% overall default rate and a negative risk-adjusted yield of -\$9.71 billion.
- **Key Risk Drivers:** Type 1 loans (unsecured credit) drive the vast majority of the \$11.70 billion in total default exposure, while Type 2 loans exhibit the highest overall default frequency at 34.54%.
- **Mandatory Underwriting Caps:** An LTV ratio greater than 107% serves as the single strongest predictor of loan default and must be instituted as a hard underwriting cap rather than a secondary risk flag.
- **Strategic Capital Reallocation:** The organization must restrict Type 1 and Type 2 originations within 'Critical' DTI bands and immediately reallocate funding toward high-yield, stable Type 3 secured debt instruments.
- **Simulated Recovery Actions:** Stress testing confirms that tighter initial underwriting protects capital far better than retroactive interest rate adjustments; enforcing a strict 60% LTV cap can reduce default exposure to \$6.37 billion and elevate the capital solvency rate to 87.06%.